Chiedozie Ihebuzor

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Education

Imperial College London **October 2022 – June 2026**

Electrical and Electronic Engineering, Current Grade: First Class Honours

- Key Modules: Digital System Design, Computer Architecture, ML, Digital VLSI Design (Apple-sponsored), Verification
- Societies: Robotics Society, Live Music Society (Guitar Technician), Skateboarding, Cheerleading, Gospel Choir.

King Edward IV College **September 2020 – May 2022**

A-Levels, Physics (A), Computer Science (A*), Maths (A*), AS Further Maths (A)*

Skills

SystemVerilog | C++ | Python | FPGAs | Cadence | Vivado | Git | Linux | Bash | Cocotb | ASICs | Arista Switches

Experience

Quantitative FPGA Technology Placement **Mar 2025 – Sept 2025**

Qube Research & Technologies *London*

- Designed a high-throughput, low-latency ML inference engine on FPGA to accelerate and up-scale existing software trading decision model.
- Created randomly generated and self-verified hardware unit tests using cocotb, and software to FPGA tests in C++/G-test, integrated into CI/CD pipeline
- Collaborated across quantitative development, research and infrastructure teams to migrate and optimise the trading model for FPGA deployment.
- Optimised C++ API to write model features to FPGA over low-latency network card efficiently.
- Parallelised team's hardware unit simulation CI with Python script, managing licences across multiple hosts and processes, reducing max runtime by 8x.

Quantitative Infrastructure Technology Intern **Jun 2024 – Sept 2024**

Qube Research & Technologies *London*

- Developed terminal-based C++ monitoring tool for L3 cache performance of trading and market access systems.
- Simplified company-wide cache policy allocation process with a terminal UI, reducing set-up errors
- Automated FPGA diagnostics logging by parsing hardware register maps and generating C++ access code via templating.

Research Intern **Aug 2023 – Sept 2023**

Imperial College London - Department of Bioengineering *London*

- Designed and developed a modular arcade platform to support game-based science outreach programs.
- Produced detailed documentation for ease of use by other researchers.

Private Tutor **Sept 2022 – Present**

Self-Employed *Remote*

- Delivered tailored tutoring in A-Level Mathematics and Physics to students to students aiming for top grades.

Projects

FPGA CORDIC Accelerator

- Implemented an unrolled, pipelined CORDIC module on an Altera FPGA to accelerate trigonometric evaluations.
- Benchmarked the module against standard C libraries to show significant speedups in calculations.

Skateboarding Data Tracker

- Wrote real-time trick detection algorithm using accelerometer, gyroscope, and ultrasonic sensor data over I^2C.
- Streamed motion analytics to Firebase and displayed data on a custom front-end dashboard.

Rubik's Cube Solver

- Implemented a solving algorithm in Python, featuring an interactive 3D visualization with real-time mouse control.
- Used OpenCV to import a user's physical cube state for automated step-by-step solving.